

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data to Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a PySpark job to load customer and order data from CSV files to Delta tables.

Target Cluster Configuration:

- Cluster Name: Data Engineering Cluster
- Databricks Runtime Version: 10.4 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 nodes, Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, pyspark

Source Data Details:

- Dataset Name: Customer Data
- Location: ``/Volumes/gen_ai_poc_databrickscOE/sdlc_wizard/customerdata``
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: Order Data
- Location: ``/Volumes/gen_ai_poc_databrickscOE/sdlc_wizard/orderdata``
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: Customer Delta Table
- Location: ``gen_ai_poc_databrickscOE.sdlc_wizard.customer``
- Format: Delta
- Description: Customer data in Delta format
- Output Name: Order Delta Table

- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order data in Delta format

Job Flow / Pipeline Stages:

- Read customer CSV data and load it to a Delta table named "customer".
- Read order CSV data and load it to a Delta table named "order".
- Add a new column "TotalAmount" to the "order" table.

Data Transformations / Business Logic:

- Step: Load customer data to Delta table
- Description: Read customer CSV data and load it to a Delta table
- Transformation Logic: Use PySpark to read CSV data and write it to a Delta table
- Step: Load order data to Delta table
- Description: Read order CSV data and load it to a Delta table
- Transformation Logic: Use PySpark to read CSV data and write it to a Delta table
- Step: Add TotalAmount column to order table
- Description: Calculate TotalAmount by multiplying PricePerUnit and Qty columns
- Transformation Logic: Use PySpark to add a new column "TotalAmount" to the "order" table

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement retry mechanism for failed tasks

TR-DLT-002: Clean and Transform Customer and Order Data

Related Functional Requirement(s): FR-DLT-002

Objective: Implement data quality checks to remove null and duplicate records from customer and order tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer data in Delta format
- Dataset Name: Order Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta

- Description: Order data in Delta format

Target Data Details:

- Output Name: Cleaned Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.customer`
- Format: Delta
- Description: Cleaned customer data in Delta format
- Output Name: Cleaned Order Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.order`
- Format: Delta
- Description: Cleaned order data in Delta format

Job Flow / Pipeline Stages:

- Remove null records from both "customer" and "order" tables.
- Remove duplicate records from both "customer" and "order" tables.

Data Transformations / Business Logic:

- Step: Remove null records
- Description: Remove null records from customer and order tables
- Transformation Logic: Use PySpark to filter out null records
- Step: Remove duplicate records
- Description: Remove duplicate records from customer and order tables
- Transformation Logic: Use PySpark to drop duplicate records

Error Handling and Logging: Same as TR-DLT-001**TR-DLT-003: Create ordersummary Table and Load Data****Related Functional Requirement(s): FR-DLT-003**

Objective: Implement a PySpark job to create an "ordersummary" table and load data into it by joining customer and order tables.

Target Cluster Configuration: Same as TR-DLT-001**Source Data Details:**

- Dataset Name: Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.customer`
- Format: Delta
- Description: Customer data in Delta format
- Dataset Name: Order Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.order`

- Format: Delta
- Description: Order data in Delta format

Target Data Details:

- Output Name: ordersummary Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.ordersummary`
- Format: Delta
- Description: ordersummary data in Delta format

Job Flow / Pipeline Stages:

- Create "ordersummary" table if it does not exist.
- Join "customer" and "order" tables on "CustId" field.
- Load the joined data into the "ordersummary" table as an SCD type 2 table.

Data Transformations / Business Logic:

- Step: Join customer and order tables
- Description: Join customer and order tables on CustId field
- Transformation Logic: Use PySpark to join the tables
- Step: Load data to ordersummary table
- Description: Load the joined data into the ordersummary table
- Transformation Logic: Use PySpark to write the data to the ordersummary table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-004: Update ordersummary Table on Customer Changes

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a PySpark job to update the "ordersummary" table whenever there is a change in the customer table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer data in Delta format

Target Data Details:

- Output Name: ordersummary Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.ordersummary`
- Format: Delta

- Description: ordersummary data in Delta format

Job Flow / Pipeline Stages:

- Identify changes in the "customer" table.
- Update the "ordersummary" table by making old records inactive and new records active.
- Update "StartDate" and "EndDate" columns accordingly to maintain history.

Data Transformations / Business Logic:

- Step: Identify changes in customer table
- Description: Identify changes in the customer table
- Transformation Logic: Use PySpark to compare the old and new customer data
- Step: Update ordersummary table
- Description: Update the ordersummary table based on the changes in customer table
- Transformation Logic: Use PySpark to update the ordersummary table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-005: Create customeraggregatespend Table and Load Data

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a PySpark job to create a "customeraggregatespend" table and load aggregated data into it.

Target Cluster Configuration: Same as TR-DLT-001**Source Data Details:**

- Dataset Name: ordersummary Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.ordersummary`
- Format: Delta
- Description: ordersummary data in Delta format

Target Data Details:

- Output Name: customeraggregatespend Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.customeraggregatespend`
- Format: Delta
- Description: customeraggregatespend data in Delta format

Job Flow / Pipeline Stages:

- Create "customeraggregatespend" table if it does not exist.
- Aggregate "TotalAmount" column from "ordersummary" table by "Name" and "Date" columns.
- Load the aggregated data into the "customeraggregatespend" table.

Data Transformations / Business Logic:

- Step: Aggregate data
- Description: Aggregate TotalAmount column by Name and Date columns
- Transformation Logic: Use PySpark to aggregate the data
- Step: Load data to customeraggregatespend table
- Description: Load the aggregated data into the customeraggregatespend table
- Transformation Logic: Use PySpark to write the data to the customeraggregatespend table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-006: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-006

Objective: Implement Delta Live Tables to execute the above steps.

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement Delta Live Tables to execute steps 1 to 5.

Data Transformations / Business Logic: Same as previous TRs

Error Handling and Logging: Same as TR-DLT-001